

# Mengjun Chen

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## Profile

Economics Ph.D. candidate specializing in macroeconomic forecasting, real-time data, and high-dimensional methods. Combines empirical macroeconomics, machine learning, and structural modeling to develop reliable economic indicators, evaluate forecasts, and interpret the effects of macroeconomic and policy shocks.

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## Research Interests

Macroeconomic Forecasting; Econometrics; Machine Learning and Deep Learning; Time Series Analysis; Macroeconomics; Monetary Policy

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## Education

### Rutgers University–New Brunswick

Ph.D. Candidate in Economics

Advisor: Norman R. Swanson

New Brunswick, NJ

Sept. 2021–Present

### National University of Singapore

M.S. in Statistics

Singapore

Aug. 2019–Jan. 2021

### Nankai University

B.A. in Finance

Tianjin, China

Mar. 2014–Jul. 2016

### Tianjin University

B.S. in Applied Chemistry

Tianjin, China

Sept. 2012–Jul. 2016

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## Research Papers

### Measuring Macroeconomic Uncertainty in Real Time: The Role of Data Revisions

*Job Market Paper*

To what extent do data revisions distort real-time measures of macroeconomic uncertainty? Using real-time data vintages, we show that data revisions persist across releases and are disproportionately concentrated during major macroeconomic disruptions. We then develop a theoretical framework that characterizes how revisions enter forecast-error uncertainty through changes in recorded outcomes and forecasts formed from historical data. The framework motivates a factor-based measure of common-state innovation uncertainty that reduces direct revision exposure by narrowing the uncertainty object. Empirically, broad factor-based measures exhibit smaller real-time distortions than their paired panel-based counterparts. Under a common VAR specification and identification scheme, both persistent-contraction and decline-and-recovery responses emerge in production and employment, with the contrast reflecting the variation retained by each uncertainty measure rather than differences in the VAR design. Forecasting comparisons also reveal economically meaningful factor-based advantages during major macroeconomic episodes. Together, these findings establish a breadth–stability tradeoff induced by data revisions: broader target-level measures preserve more forecast-error variation but carry greater revision exposure, whereas common-state measures limit real-time distortion by narrowing the scope of uncertainty.

### Learning Forecastable Factors When Targets Are Scarce

*Work in Progress*

We develop a deep learning framework for recovering forecastable latent structure from high-dimensional economic panels when target supervision is scarce. By jointly learning nonlinear representations and recurrent dynamics, the framework transforms the panel’s future evolution into temporal supervision, isolating predictive common variation from dominant but transitory fluctuations. Our approach bridges modern representation learning with classical dynamic factor methods and nests reduced-rank forecasting as a linear special case.

**Labor Market Segmentation, Global Sourcing, and Macroeconomic Adjustment**

*Work in Progress*

This paper examines how labor-market segmentation shapes the endogenous organization of global production. We develop a dynamic small-open-economy model with heterogeneous labor pools that differ in matching frictions and their positions within the production process. Scarcity in one pool induces greater reliance on imported inputs while preserving domestic production, whereas scarcity in another shifts sourcing toward foreign final goods and contracts domestic value added more broadly. Consequently, aggregate labor scarcity is not a sufficient statistic for sourcing, employment, or output dynamics. By feeding back into vacancy creation and domestic productive capacity, global sourcing can amplify and perpetuate transitory labor-market disruptions. The framework delivers new implications for trade policy, occupational training, hiring subsidies, and skill-targeted immigration.

**Software and Research Projects**

**Developer, avgEffect R Package** Completed Oct. 2025

- Developed an R package for coefficient and average partial-effect estimation with heteroskedasticity-robust and one- and two-way clustered standard errors.

**Professional Experience**

**Data Strategy Specialist Intern (Full-time), Baidu, Inc.** China  
*Data Strategy and Analytics* Mar. 2021–Jun. 2021

- Contributed to the establishment of a SQL-based data analysis system.

**Teaching Experience**

**Rutgers University–New Brunswick** New Brunswick, NJ  
*Lecturer, Intermediate Macroeconomic Analysis (Econ 321)* Summers 2024 and 2025

- Independently taught undergraduate intermediate macroeconomics.

*Teaching Assistant* 2022–Present

- Graduate courses: Macroeconomic Theory II; Econometrics I.
- Undergraduate courses: Introduction to Microeconomics; Introduction to Macroeconomics; Intermediate Macroeconomic Analysis; Money, Banking and Finance; Econometrics.

**Presentation**

- “Labor Market Segmentation, Global Sourcing, and Macroeconomic Adjustment,” Chinese Economists Society (CES) North American Annual Conference, 2025.

**Honors and Awards**

- Professor Hiroki Tsurumi and Peter J. Zangari Endowed Graduate Dissertation Fund in Economics for Exceptional Dissertation Research, 2026.
- Sidney I. Simon Prize for Outstanding Graduate Student Teaching, 2025.
- Richard Lock Endowed Fund for Economics Award, 2025.
- Rie Ashizawa Memorial Award/Prize, 2022.

**Skills and Languages**

**Programming and Data:** MATLAB, R, Python, SQL (MySQL)

**Econometrics, Forecasting, and Machine Learning:** Time-series econometrics, factor models, VARs, autoencoders, RNNs, and LSTMs

**Structural Modeling:** Dynare; DSGE model solution, calibration, and estimation

**Languages:** Mandarin (native), English (fluent)